

# Multiple Discrete Random Variables

## Lecture 5

# Topics in this lecture

- Joint PMF
- Conditional PMF
- Conditional expectation
- Independence of random variables

# Joint PMF

- Let  $X$  and  $Y$  be discrete random variables, the joint PMF of  $X$  and  $Y$ , denoted by  $p_{X,Y}(x,y)$ , captures the probabilities of the events in the form of  $\{X = x, Y = y\}$
- Joint probability mass function

$$p_{X,Y}(x, y) = P(X = x, Y = y)$$

# Compute Probability from Joint PMF

- Finding probability of event  $A$

$$P(A) = \sum_{(x,y) \in A} p_{X,Y}(x,y)$$

- Sum over all values of joint pmf

$$\sum_{\forall(x,y)} p_{X,Y}(x,y) = 1$$

# Example 5.1

- Let  $X$  and  $Y$  be the marks that show up after rolling two 4-faced dice. The joint PMF  $p_{X,Y}(x,y)$  is given in the following table. Find the probability of an event  $\{X > 3, Y < 3\}$

|     |   | $X$  |      |      |      |
|-----|---|------|------|------|------|
|     |   | 1    | 2    | 3    | 4    |
| $Y$ | 1 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 2 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 3 | 1/24 | 1/24 | 1/24 | 1/24 |
|     | 4 | 2/24 | 2/24 | 2/24 | 2/24 |

# Example

- A statistic shows that 10% of people entering a store will buy an iPhone, 30% will buy an iPad, and 60% will not buy anything. If 10 people enter the store today, what is the probability that exactly 2 iPhone and 2 iPad will be sold?

100%

iPhone 10%  
 iPad 30%  
 Nothing 60%

0.01  
 0.03

0.04

# Marginal PMF

$$\begin{aligned} p_X(x) &= P(X = x) \\ &= P(X = x, Y = \text{anything}) \\ &= \sum_y p_{X,Y}(x, y) \end{aligned}$$

Similarly

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

# Example 5.2

- Let  $X$  and  $Y$  be the marks that show up after rolling two 4-faced dice. The joint PMF  $p_{X,Y}(x,y)$  is given in the following table. Find the marginal PMF of  $X$ .

|     |   | $X$  |      |      |      |
|-----|---|------|------|------|------|
|     |   | 1    | 2    | 3    | 4    |
| $Y$ | 1 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 2 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 3 | 1/24 | 1/24 | 1/24 | 1/24 |
|     | 4 | 2/24 | 2/24 | 2/24 | 2/24 |

# Conditional PMF

- Conditional PMF on Event A

$$p_X(x|A) = P(X = x|A) = \frac{P(\{X = x\} \cap A)}{P(A)}, \quad P(A) > 0$$

- Conditional PMF

$$p_{X|Y}(x|y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{p_{X,Y}(x, y)}{p_Y(y)}$$

# Example 5.3

- Let  $X$  and  $Y$  be the marks that show up after rolling two 4-faced dice. The joint PMF  $p_{X,Y}(x,y)$  is given in the following table. Find the conditional PMF of  $X|Y=2$ .

|     |   | $X$  |      |      |      |
|-----|---|------|------|------|------|
|     |   | 1    | 2    | 3    | 4    |
| $Y$ | 1 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 2 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 3 | 1/24 | 1/24 | 1/24 | 1/24 |
|     | 4 | 2/24 | 2/24 | 2/24 | 2/24 |

# Example

- Pick a number  $X$  at random from a set  $\{1, 2, 3, 4, 5\}$ . Then, pick another random number  $Y$ , where  $Y$  is a positive integer that is less than or equal to  $X$ .
  - Find the joint PMF of  $X, Y$
  - Find the conditional PMF of  $X \mid Y = 3$

# Conditional Expectation

- Conditional expectation of  $X|Y = y$

$$E[X|Y = y] = \sum_x x p_{X|Y}(x|y)$$

- Finding  $E[X]$  from conditional expectation

$$E[X] = E[E[X|Y]] = \sum_y E[X|Y = y] p_Y(y)$$

# Example 5.4

- Let  $X$  and  $Y$  be the marks that show up after rolling two 4-faced dice. The joint PMF  $p_{X,Y}(x,y)$  is given in the following table. Find the conditional expectation  $E[X|Y=2]$ .

|     |          |          |          |          |          |
|-----|----------|----------|----------|----------|----------|
|     |          | $X$      |          |          |          |
|     |          | <b>1</b> | <b>2</b> | <b>3</b> | <b>4</b> |
| $Y$ | <b>1</b> | 1/24     | 2/24     | 2/24     | 1/24     |
|     | <b>2</b> | 1/24     | 2/24     | 2/24     | 1/24     |
|     | <b>3</b> | 1/24     | 1/24     | 1/24     | 1/24     |
|     | <b>4</b> | 2/24     | 2/24     | 2/24     | 2/24     |

# Example

- Pick a number  $X$  at random from a set  $\{1, 2, 3, 4, 5\}$ . Then, pick another random number  $Y$ , where  $Y$  is a positive integer that is less than or equal to  $X$ .
- Find the conditional expectation  $E[X \mid Y = 3]$

# Independence of 2 RV

- If  $X$  and  $Y$  are independent discrete random variables, then the joint pmf is equal to the product of the marginal pmf

$$p_{X,Y}(x,y) = p_X(x)p_Y(y) \quad , \quad \forall x,y$$

# Example 5.5

- Let  $X$  and  $Y$  be the marks that show up after rolling two 4-faced dice. The joint PMF  $p_{X,Y}(x,y)$  is given in the following table. Are  $X, Y$  independent?

|     |   | $X$  |      |      |      |
|-----|---|------|------|------|------|
|     |   | 1    | 2    | 3    | 4    |
| $Y$ | 1 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 2 | 1/24 | 2/24 | 2/24 | 1/24 |
|     | 3 | 1/24 | 1/24 | 1/24 | 1/24 |
|     | 4 | 2/24 | 2/24 | 2/24 | 2/24 |